

E-Commerce Customer Analytics

Production SQL Portfolio & Business Strategy Matrix

Analysis Scope & Context

End-to-end customer portfolio assessment covering 3,900+ analytical event transactions. Designed to isolate highest-performing cohorts, benchmark promotional campaign efficacy, evaluate subscription behavior, and optimize customer lifetime value (CLV).

1. Total Revenue Contribution by Gender

```
SELECT
  gender,
  SUM(purchase_amount) AS total_revenue
FROM mydata.customer
GROUP BY gender
ORDER BY total_revenue DESC;
```

BUSINESS OBJECTIVE

Evaluate top-line demographic volume splits across product lines.

STRATEGIC APPLICATION

Guides high-level promotional allocations and programmatic lookalike audience acquisition sizing.

2. Identification of Top 10 High-Spending Customers

```
SELECT
  customer_id,
  SUM(purchase_amount) AS total_spent
FROM mydata.customer
GROUP BY customer_id
ORDER BY total_spent DESC
LIMIT 10;
```

BUSINESS OBJECTIVE

Isolate structural VIP users driving a disproportionate percentage of store sales.

STRATEGIC APPLICATION

Feeds direct customer success white-glove touchpoints and exclusive access groups.

3. Aggregated Sales Revenue by Product Category

```
SELECT
    category,
    SUM(purchase_amount) AS total_sales
FROM mydata.customer
GROUP BY category
ORDER BY total_sales DESC;
```

BUSINESS OBJECTIVE

Profile the primary revenue engines of the enterprise storefront catalog.

STRATEGIC APPLICATION

Informs cross-departmental supply chain capacity sizing and seasonal warehouse stocking rules.

4. Customer Satisfaction Analysis via Average Rating by Category

```
SELECT
    category,
    ROUND(AVG(review_rating), 2) AS avg_rating
FROM mydata.customer
GROUP BY category
ORDER BY avg_rating DESC;
```

BUSINESS OBJECTIVE

Uncover explicit category quality issues or high customer sentiment nodes.

STRATEGIC APPLICATION

Flags lagging product categories to the quality assurance or merchant relations vertical.

5. Seasonal Macro Sales and Order Volume Analysis

```
SELECT
    season,
    SUM(purchase_amount) AS total_sales,
    COUNT(*) AS total_orders
FROM mydata.customer
GROUP BY season
ORDER BY total_sales DESC;
```

BUSINESS OBJECTIVE

Track calendar macro-shifts and transaction velocity changes throughout the fiscal year.

STRATEGIC APPLICATION

Optimizes capital allocation budgets for paid media across peak vs off-peak cycles.

6. Payment Method Penetration and Revenue Generation

```
SELECT
    payment_method,
    COUNT(*) AS total_transactions,
    SUM(purchase_amount) AS revenue
FROM mydata.customer
GROUP BY payment_method
ORDER BY revenue DESC;
```

BUSINESS OBJECTIVE

Map out check-out rail engagement and identify vendor transactional volume.

STRATEGIC APPLICATION

Provides leverage for interchange merchant fee contract negotiations with processors.

7. Subscription Program Efficacy and Cohort Values

```
SELECT
    subscription_status,
    COUNT(*) AS customers,
    SUM(purchase_amount) AS total_sales,
    ROUND(AVG(purchase_amount), 2) AS avg_purchase
FROM mydata.customer
GROUP BY subscription_status;
```

BUSINESS OBJECTIVE

Quantify the baseline variance in checkout behavior between members and non-members.

STRATEGIC APPLICATION

Validates CAC (Customer Acquisition Cost) allowances for recurring membership models.

8. Promotional Discount Impact on Basket Size Metrics

```
SELECT
    discount_applied,
    SUM(purchase_amount) AS total_sales,
    ROUND(AVG(purchase_amount), 2) AS avg_purchase
FROM mydata.customer
GROUP BY discount_applied;
```

BUSINESS OBJECTIVE

Determine whether coupon campaigns dilution reduces net margin or expands order volume.

STRATEGIC APPLICATION

Assists the pricing division in assessing coupon dependency behaviors.

9. Top 10 Most Frequently Purchased Products

```
SELECT
    item_purchased,
    COUNT(*) AS total_orders
FROM mydata.customer
GROUP BY item_purchased
ORDER BY total_orders DESC
LIMIT 10;
```

BUSINESS OBJECTIVE

Map transaction velocities at individual SKU levels.

STRATEGIC APPLICATION

Informs immediate front-page site real-estate layouts and cross-sell recommendation engines.

10. Top 10 Regional Revenue Generating Locations

```
SELECT
    location,
    SUM(purchase_amount) AS total_sales
FROM mydata.customer
GROUP BY location
ORDER BY total_sales DESC
LIMIT 10;
```

BUSINESS OBJECTIVE

Locate major geographical demand clusters across physical territories.

STRATEGIC APPLICATION

Guides localization strategies and physical warehouse distribution node positioning.

11. Age Demographic Spending Profile Analysis

```
SELECT
    age_group,
    COUNT(*) AS customers,
    SUM(purchase_amount) AS total_sales,
    ROUND(AVG(purchase_amount), 2) AS avg_spending
FROM mydata.customer
GROUP BY age_group
ORDER BY total_sales DESC;
```

BUSINESS OBJECTIVE

Identify primary age segments responsible for the business's unit volume.

STRATEGIC APPLICATION

Directly optimizes brand creative voice choices and channel target selections.

12. Shipping Tier Performance and Associated Revenue

```
SELECT
    shipping_type,
    COUNT(*) AS total_orders,
    SUM(purchase_amount) AS revenue
FROM mydata.customer
GROUP BY shipping_type
ORDER BY revenue DESC;
```

BUSINESS OBJECTIVE

Measure user preference regarding delivery speeds and cost configurations.

STRATEGIC APPLICATION

Supports third-party logistics (3PL) negotiations for dedicated fast-shipping tier structures.

13. Identification of High-Loyalty Customer Bases

```
SELECT
    customer_id,
    previous_purchases,
    purchase_amount
FROM mydata.customer
WHERE previous_purchases >= 20
ORDER BY previous_purchases DESC;
```

BUSINESS OBJECTIVE

Filter out the core repetitive historical purchasing cluster.

STRATEGIC APPLICATION

Feeds persistent audience lists for targeted retention and brand ambassador programs.

14. Average Ticket Values across Specific Product Categories

```
SELECT
    category,
    ROUND(AVG(purchase_amount), 2) AS avg_purchase
FROM mydata.customer
GROUP BY category
ORDER BY avg_purchase DESC;
```

BUSINESS OBJECTIVE

Differentiate low-ticket/high-volume categories from high-ticket segments.

STRATEGIC APPLICATION

Establishes category-specific thresholds for free shipping promotions.

15. Multi-Tier Value Customer Segmentation using CASE

```
SELECT
  customer_id,
  purchase_amount,
  CASE
    WHEN purchase_amount >= 80 THEN 'High Value'
    WHEN purchase_amount >= 50 THEN 'Medium Value'
    ELSE 'Low Value'
  END AS customer_segment
FROM mydata.customer;
```

BUSINESS OBJECTIVE

Group individual order footprints dynamically into clean operational strata.

STRATEGIC APPLICATION

Enables conditional cross-selling rules directly within marketing automation tools.

16. Revenue Share Contribution Percentages via Analytic Subqueries

```
SELECT
  category,
  SUM(purchase_amount) AS total_sales,
  ROUND(
    SUM(purchase_amount) * 100.0 /
    (SELECT SUM(purchase_amount) FROM mydata.customer),
    2
  ) AS revenue_percentage
FROM mydata.customer
GROUP BY category
ORDER BY total_sales DESC;
```

BUSINESS OBJECTIVE

Expose dependency profiles of the overall business on specific dominant categories.

STRATEGIC APPLICATION

Assists corporate planners in monitoring risk concentration in the product catalog.

17. Windowed Ranking Assignments Across Customer Bases

```
SELECT
  customer_id,
  SUM(purchase_amount) AS total_spent,
  RANK() OVER (
    ORDER BY SUM(purchase_amount) DESC
  ) AS spending_rank
FROM mydata.customer
GROUP BY customer_id;
```

BUSINESS OBJECTIVE

Build a strict cumulative leaderboard of single customer spend metrics.

STRATEGIC APPLICATION

Powers tier assignment operations within external loyalty software applications.

18. Cumulative Windowed Running Totals Tracking

```
SELECT
  customer_id,
  purchase_amount,
  SUM(purchase_amount) OVER (
    ORDER BY customer_id
  ) AS running_revenue
FROM mydata.customer;
```

BUSINESS OBJECTIVE

Track mathematical transaction progressions sequentially across transaction records.

STRATEGIC APPLICATION

Feeds localized charting software applications to visualize transaction velocity paths.

19. Individual Values Compared Against Global Historical Average

```
SELECT
  customer_id,
  purchase_amount,
  (SELECT AVG(purchase_amount) FROM mydata.customer) AS overall_avg,
  purchase_amount - (SELECT AVG(purchase_amount) FROM mydata.customer) AS difference
FROM mydata.customer;
```

BUSINESS OBJECTIVE

Quantify order-level deviation against macro benchmarks.

STRATEGIC APPLICATION

Assists real-time fraud or anomaly detection engines in flagging extreme purchase behaviors.

20. Top Category Demands Grouped and Ordered by Gender Split

```
SELECT
  gender,
  category,
  SUM(purchase_amount) AS total_sales
FROM mydata.customer
GROUP BY gender, category
ORDER BY gender, total_sales DESC;
```

BUSINESS OBJECTIVE

Understand how catalog preference indices shift across specific gender classes.

STRATEGIC APPLICATION

Optimizes email product recommendation arrays based on customer demographic flags.

21. Multi-Strata Loyalty Index Classification Matrix

```
SELECT
  customer_id,
  previous_purchases,
  CASE
    WHEN previous_purchases >= 40 THEN 'Highly Loyal'
    WHEN previous_purchases >= 20 THEN 'Loyal'
    WHEN previous_purchases >= 10 THEN 'Regular'
    ELSE 'New'
  END AS loyalty_segment
FROM mydata.customer;
```

BUSINESS OBJECTIVE

Categorize your user base by frequency and maturity of lifecycle touchpoints.

STRATEGIC APPLICATION

Triggers specialized win-back discount rules for lagging 'Regular' or 'Loyal' tiers.

22. Seasonal Performance Trends per Individual Category Segment

```
SELECT
  category,
  season,
  SUM(purchase_amount) AS revenue
FROM mydata.customer
GROUP BY category, season
ORDER BY category, revenue DESC;
```

BUSINESS OBJECTIVE

Isolate cross-dependencies between times of year and functional product groupings.

STRATEGIC APPLICATION

Prevents off-season inventory overstocking across highly volatile fast-moving lines.

23. Transaction Shares Across Alternative Checkout Configurations

```
SELECT
  payment_method,
  COUNT(*) AS transactions,
  ROUND(
    COUNT(*) * 100.0 / (SELECT COUNT(*) FROM mydata.customer),
    2
  ) AS transaction_percentage
FROM mydata.customer
GROUP BY payment_method;
```

BUSINESS OBJECTIVE

Expose user interface payment configuration popularity breakdowns.

STRATEGIC APPLICATION

Helps prioritize product team bandwidth for optimizing alternative checkout payment rails.

24. Operational Split: First-Time vs Repeat Purchases

```
SELECT
  customer_id,
  previous_purchases,
  CASE
    WHEN previous_purchases > 1 THEN 'Repeat Customer'
    ELSE 'First-Time Customer'
  END AS purchase_type
FROM mydata.customer;
```

BUSINESS OBJECTIVE

Isolate new client acquisitions from the core repeat customer engine.

STRATEGIC APPLICATION

Monitors structural health and customer retention sustainability metrics.

25. Top Rated Categories Subject to Minimum Revenue Floor

```
SELECT
  category,
  ROUND(AVG(review_rating), 2) AS avg_rating,
  SUM(purchase_amount) AS total_sales
FROM mydata.customer
GROUP BY category
HAVING SUM(purchase_amount) > 5000
ORDER BY avg_rating DESC;
```

BUSINESS OBJECTIVE

Surface high-sentiment categories while filtering out low-volume statistical anomalies.

STRATEGIC APPLICATION

Identifies viable, high-quality candidates for inclusion in main banner advertisement spots.

26. Operational Purchase Distributions across Age Demographics

```
SELECT
  age_group,
  MIN(purchase_amount) AS min_purchase,
  MAX(purchase_amount) AS max_purchase,
  ROUND(AVG(purchase_amount), 2) AS avg_purchase
FROM mydata.customer
GROUP BY age_group;
```

BUSINESS OBJECTIVE

Map structural variance, spread bounds, and price sensitivities across ages.

STRATEGIC APPLICATION

Enables product merchandising units to adjust catalog tier options based on age profile limits.

27. Dominant High-Revenue Product Identification inside Categories

```
SELECT
    category,
    item_purchased,
    SUM(purchase_amount) AS revenue
FROM mydata.customer
GROUP BY category, item_purchased
ORDER BY category, revenue DESC;
```

BUSINESS OBJECTIVE

Perform granular product-line assessment to isolate category flagships.

STRATEGIC APPLICATION

Informs supplier contracts and optimizes marketing asset generation priorities.

28. Cross-Tabulated Coupon/Discount Sensitivity by Gender Split

```
SELECT
    gender,
    discount_applied,
    COUNT(*) AS total_customers
FROM mydata.customer
GROUP BY gender, discount_applied;
```

BUSINESS OBJECTIVE

Uncover pricing elasticities and coupon affinity variances across gender lines.

STRATEGIC APPLICATION

Tailors promotional copywriting configurations to match identified cohort expectations.

29. Revenue Value Distributions across Stated Purchase Frequencies

```
SELECT
    frequency_of_purchases,
    SUM(purchase_amount) AS total_sales,
    ROUND(AVG(purchase_amount), 2) AS avg_purchase
FROM mydata.customer
GROUP BY frequency_of_purchases
ORDER BY total_sales DESC;
```

BUSINESS OBJECTIVE

Validate if self-declared or systemic frequencies align with higher spending metrics.

STRATEGIC APPLICATION

Aids lifecycle marketing in determining optimal message pacing and content scheduling parameters.

30. First-Order Approximation Model for Customer Lifetime Value (CLV)

```
SELECT
  customer_id,
  purchase_amount * previous_purchases AS estimated_clv
FROM mydata.customer
ORDER BY estimated_clv DESC;
```

BUSINESS OBJECTIVE

Model systemic proxy metrics for customer historical equity values.

STRATEGIC APPLICATION

Establishes baseline values used to cross-verify channel-level acquisition yield targets.